

$$\hbar = c = 1$$

$$L = CT$$

$$c = \lambda \omega$$

$$p = \frac{\hbar}{\lambda} = mc \Rightarrow$$

$$L = \hbar m^{-1} c^{-1}$$

$$[m] = +1$$

$$[x^\mu] = -1$$

$$[\partial^\mu] = +1$$

$$[d^d x] = -d$$

$$\leftarrow \frac{\partial}{\partial x^\mu} \sim \frac{\partial}{\partial (\hbar m^{-1} c^{-1})} \sim \frac{\partial}{\partial r} m$$

$$\mathcal{L} = -\frac{1}{2} \partial^\mu \varphi \partial_\mu \varphi \sim \frac{1}{2} m^2 \varphi - \sum_{n=3}^{\infty} \frac{1}{n!} g_n \varphi^n$$

$$S = \int d^d x \mathcal{L}$$

$$Z(\mathcal{J}) = \int \mathcal{D}\varphi e^{i \int d^d x (\mathcal{L} + \mathcal{J}\varphi)}$$

$$[S] = 0 \Rightarrow [\mathcal{L}] = d$$

$$[-\frac{1}{2} m^2 \varphi^2] = d = 2 + 2[\varphi]$$

$$2[\varphi] = d - 2 \Rightarrow [\varphi] = \frac{1}{2}(d - 2)$$

$$[g_n \varphi^n] = [g_n] + \frac{n}{2}(d - 2) = d$$

$$[g_n] = d - \frac{n}{2}(d - 2)$$

* For ϕ^3 : $n = 3$

$$[g_3] = d - \frac{3}{2}d + 3 = 3 - \frac{d}{2} = \frac{1}{2}(6 - d)$$

• Coupling const. of ϕ^3 is dimensionless in $d = 6$

$g \phi^{3/2}$: the right coupling

→ Will look at ϕ^3 in $d = 6$ where $[g_3] = 0$